

IB MATH AA SL

Lesson 29

Conditional Probability & Trees

Statistics & Probability · 60-minute lesson

Syllabus SL 4.6 · Property of Lynda Badmus Education

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Do Now — recall

ON YOUR WHITEBOARD

1. Two coins: $P(\text{two heads})$?
2. Does drawing without replacement change later probabilities?
3. $P(6 \text{ then } 6)$ on two dice?

Answers (click to reveal)

1. $\frac{1}{4}$
2. yes
3. $\frac{1}{36}$

What we're learning today

LEARNING OBJECTIVES

- 1 Draw**
and use tree diagrams.
- 2 Multiply**
along branches; add between paths.
- 3 Use**
conditional probability.
- 4 Handle**
with/without replacement.

SUCCESS CRITERIA — I CAN...

- ✓
build a tree diagram for two stages
- ✓
multiply along a branch path
- ✓
add the relevant paths for a combined event
- ✓
apply the conditional probability formula

Tree diagrams

AND $\times$ • OR $+$

$$P(\text{path}) = P_1 \times P_2$$

Along branches

MULTIPLY

Between paths

ADD

Check

probabilities on each set of branches sum to 1

REPLACEMENT

With

probabilities stay the same

Without

the total (and counts) change on the 2nd draw

Tree probability

QUESTION

A bag has 3 red, 2 blue. Two balls are drawn WITHOUT replacement.

Find $P(\text{both red})$.

SOLUTION

1st red: $\frac{3}{5}$. Then 2 red left of 4:

$$P(\text{both red}) = \frac{3}{5} \times \frac{2}{4}$$

Multiply along the branch:

$$= \frac{6}{20} = \frac{3}{10}$$

Your turn — trees

ANSWER IN YOUR BOOK

1. P(two heads) on two fair coins.
2. Bag 4 red, 1 blue (no replacement):
P(red then blue)?
3. P(tail then 6) coin & die?

Answers (click to reveal)

1. $\frac{1}{4}$
2. $\frac{4}{5} \times \frac{1}{4} = \frac{1}{5}$
3. $\frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

Conditional probability

GIVEN THAT...

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

P(A | B)

probability of A GIVEN B has happened

Divide

the overlap by the condition's probability

INDEPENDENCE

Independent if $P(A \cap B) = P(A) \cdot P(B)$

Then

$P(A | B) = P(A)$ — B makes no difference.

Conditional probability

QUESTION

$P(A \cap B) = 0.12$ and $P(B) = 0.3$.

Find $P(A \mid B)$.

SOLUTION

Use the conditional formula:

$$P(A \mid B) = \frac{0.12}{0.3}$$

Evaluate:

$$= 0.4$$

Exam-style question

A box has 5 good and 3 faulty bulbs. Two are taken without replacement.

(a) Find $P(\text{both good})$. [3]

(b) Find $P(\text{exactly one faulty})$. [3]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $5/8 \times 4/7 = 20/56 = 5/14$ (M1)(A1)(A1)

(b) $GF + FG = 5/8 \cdot 3/7 + 3/8 \cdot 5/7 = 30/56 = 15/28$ (M1)(A1)(A1)

Common mistakes



And/Or

Multiply along a path (and); add across paths (or).



Without replacement

Reduce the count AND the total on the 2nd draw.



Conditional formula

Divide the intersection by the GIVEN event.



Branches sum to 1

Check each split of branches.

Mixed practice

QUESTIONS

1. $P(\text{two sixes})$ on two dice.
2. Bag 2 red, 3 blue (no repl.): $P(\text{both blue})$?
3. $P(A \cap B) = 0.2$, $P(B) = 0.5$: $P(A|B)$?
4. Independent A, B : $P(A) = 0.4$, $P(B) = 0.5$: $P(A \cap B)$?

Answers (click to reveal)

1. $1/36$
2. $3/5 \times 2/4 = 3/10$
3. 0.4
4. 0.2

Plenary & exit ticket

KEY FORMULAS

$$P(\text{path}) = P_1 \times P_2$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

EXIT TICKET — before you leave

1. Along branches: $\times$ or $+$?
2. $P(A \cap B) = 0.1, P(B) = 0.4$: $P(A | B)$?
3. Two coins: $P(\text{two tails})$?

Answers (click to reveal)

1. $\times$
2. 0.25
3. $1/4$

Self-assess:
 • need more help
 • nearly there
 • confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Conditional probability & trees.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 30

Discrete random variables and expected value.

$$E(X) = \sum x P(X=x)$$